Activity – 17

Writing a menu driven choice based c program which takes input from the user and do following operations.

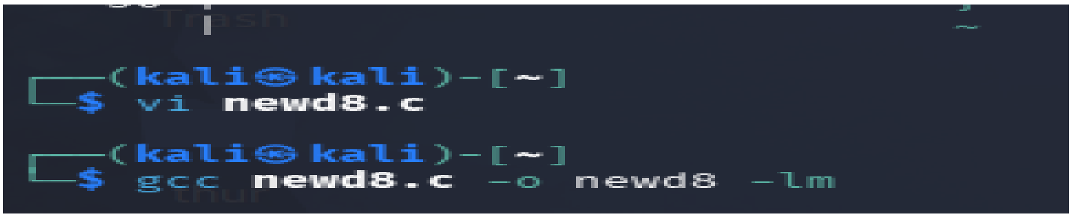
1. Making the largest digit out of the number.
2. Making the smallest digit out of the number.
3. Swapping the last and first digit of the number.
4. Checking if the number is a palindrome.
5. Exit

Program

```
#include <stdio.h>
#include <string.h>


void largestnumber();
void smallestnumber();
void swapfirstlast();
void checkpalindrome();


int main() {
        int choice;
        do{
                printf("\n ——Menu——\n");
                printf("\n first choice is for generate largest number \n");
                printf("\n Second choice is for generate smallest number \n");
                printf("\n Third choice is for swapping first and last  digits \n");
                printf("\n fourth choice is for checking palindrome \n");
                printf("\n fifth choice is for exiting \n");
                printf("Enter a choice: ");
                scanf("%d", &choice);

                switch(choice){
                        case 1: largestnumber();  break;
                        case 2: smallestnumber(); break;
                        case 3: swapfirstlast(); break;
                        case 4: checkpalindrome(); break;
                        case 5: printf("We are exiting"); break;
                        default: printf("Invalid choice.\n");
                        }
        }while(choice ≠ 5);
        return 0;
}

void largestnumber(){
        int num;
        printf("Enter the number: ");
        scanf("%d", &num);
        char str[20];
        sprintf(str, "%d", num);
        int len = strlen(str);
        for(int i=0; i<len-1; i++){
                for(int j=i+1; j<len; j++){
                        if(str[i] < str[j]){
                                char temp = str[i];
                                str[i] = str[j];
                                str[j] = temp;
                        }
                }
        }
```

```c
	printf("Largest number: %s\n", str);
}

void smallestnumber(){
	int num;
	printf(" number: ");
	scanf("%d", &num);
	char str[20];
	sprintf(str, "%d", num);
	int len = strlen(str);
	for(int i = 0; i<len - 1; i++){
		for(int j=i+1; j<len; j++){
			if(str[i]>str[j]){
				char temp=str[i];
				str[i]=str[j];
				str[j]=temp;
			}
		}
	}
	printf("Smallest digit: %s\n", str);
}
void swapfirstlast(){
	int num;
	printf("Enter a positive integer.\n");
	scanf("%d", &num);
	char str[20];
	sprintf(str, "%d", num);
	int len=strlen(str);
	char temp=str[0];
	str[0] = str[len - 1];
	str[len - 1] = temp;
	printf("After swapping: %s\n", str);
}

void checkpalindrome(){
	int num;
	printf("enter a positive integer: ");
	scanf("%d", &num);
	char str[20], rev[20];
	sprintf(str, "%d", num);
	int len = strlen(str);
	for(int i = 0; i<len; i++){
		rev[i] = str[len - 1 - i];
	}
	rev[len] = '\0';
	if(strcmp(str, rev)==0)
		printf("palindrome number.\n");
	else
		printf("not a palindrome number");
}
```

Compilation and execution



Algorithm

1.Start

2.Input a positive integer from user.

3.Menu display with all our five choices.

4.Read choice.

5. use switch builtin to make cases for each choice.

6. Make void functions to be used in switch cases for each operation.

7.STOP

Test case table

Input	choice	Output	Expected output	Result
12343	1	43321	43321	PASS
32122	2	12223	12223	PASS
14223	3	34221	34221	PASS
13121	4	no	no	PASS